OHM Campus Challenge 2026 - Bajaj Auto

***Title*: System-Level Proposal for Integrating Next-Generation Features, Fast Charging, and TPMS in a Commercial Three-Wheeler EV**

Team Name: Current Crowns
Team Members:
- Jennita Rachel A
- Kokkonda Varshha

College: Indian Institute of Technology Guwahati
Academic Background: Currently enrolled in the second year of the Bachelor of Technology program, specializing in Engineering Physics.

- **Next gen features:**

Theft prevention alert: This can be achieved through biometric system
The fingerprint system uses the existing microcontroller, dashboard, and GPS unit, so no extra major hardware is required.
Vehicle security is achieved through software-based motor immobilization instead of costly mechanical locks.
Unauthorized access attempts trigger a GPS alert to the fleet manager.
A simple PIN backup ensures reliability if fingerprint detection fails.
By reusing existing systems and minimizing added components, this solution remains affordable and practical for commercial fleets.

Vehicle Use Case & Operating Assumptions:

Vehicle Purpose: The proposed vehicle is a commercial electric three-wheeler goods carrier made for logistics and urban cargo transport. The design prioritizes high uptime, thermal robustness, cost effectiveness, and durability under harsh operating conditions.

1.2 Operating Assumptions:

Parameter	**Assumed Range**	**Justification**
Daily Running	100-150 km/day	Typical urban last-mile delivery routes
Operating hours	10-14 hrs/day	Commercial multi-shift usage
Payload	300 - 500 kg	Standard cargo category
Average speed	Nearly 25kmph	Congested urban traffic
Ambient conditions	High temperature,humidity, rain,dust,flooding	Indian operating environment
User Type	Commercial driver	Focus on uptime & reliability

2 Energy Consumption Estimation:

2.1 Physics-Based Estimation

Forces Acting on the Vehicle (a) Rolling Resistance

$F_{rr} = C_{rr} \cdot m \cdot g$ Where: $C_{rr} = 0.015–0.02$ (commercial tyres)

m = total mass (vehicle + payload)

g = 9.81 m/s2

Assuming: • Vehicle mass ≈ 450 kg •

Payload ≈ 450 kg Total mass ≈ 900 kg

Crr = 0.018 Frr = 0.018 × 900 × 9.81 ≈ 159

Energy per km (rolling only): 159 × 1000 = 159,000 J ≈ 44Wh/km

Aerodynamic Drag , Fd = 1 2 ρCdAv

For a 3-wheeler: CdA ≈ 1.2–1.6 m2 v ≈ 25–40 km/h

At urban speeds, aerodynamic drag is secondary compared to rolling resistance.

Total Energy Consumption

- Rolling resistance: ~44 Wh/km
- Aerodynamic + stop-go losses: ~30–40 Wh/km •
- Electrical & drivetrain losses: ~10–15 Wh/km

Total ≈ 80–100 Wh/km A conservative design value of 90 Wh/km is therefore selected.

2.2 Sanity Check (Back-Calculation)

Energy per km = 10,000Wh

110 km ≈ 91Wh/km

This aligns with both physics-based and field data estimates.

3 Battery Capacity Selection

3.1 Battery Chemistry Selection Lithium Iron Phosphate (LFP) is selected.

Justification

- Very high thermal stability
- Thermal runaway temperature ≈ 270–300 ◦C
- Long cycle life: 3000–6000 cycles (8–10 years commercial use)
- Tolerates frequent partial charging
- Lower degradation under heat stress
- Cost-effective for commercial vehicles

3.2 Battery Capacity Derivation(Assuming):

- Daily distance = 120 km
- Energy consumption = 90 Wh/km Daily Energy = 120 × 90 = 10.8 kWh Including buffer for: Thermal stress , Traffic congestion, Battery aging ,Auxiliary loads

Buffer ≈ 25–30% **Final Battery Capacity ≈ 14 kWh**

4 Power Consumption (Average Operating Power)

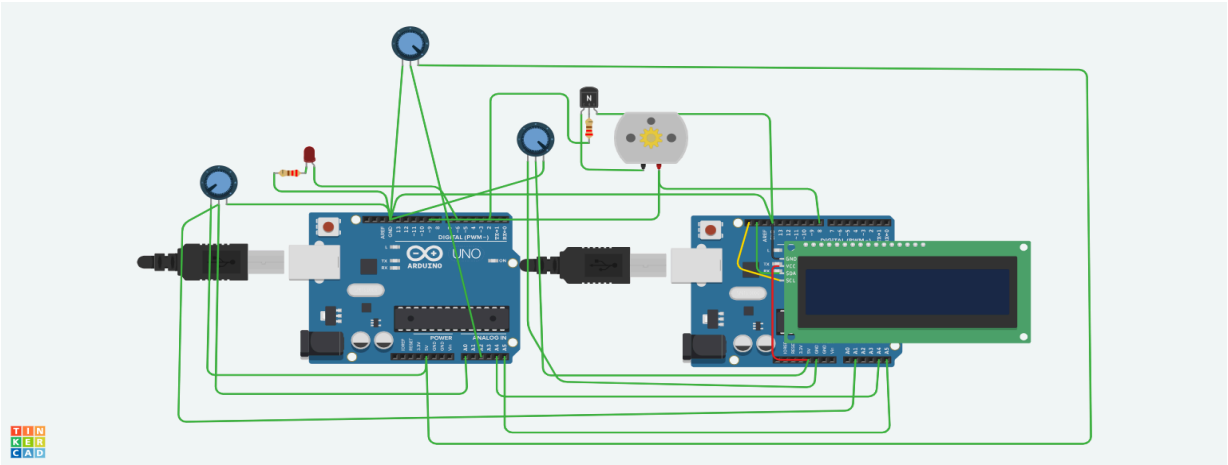
Power consumption indicates how fast energy is used, Assuming: • Average speed ≈ 25 km/h • 100 km covered in ≈ 4 hours **Average Power = 11 kWh 4 h ≈ 2.75 kW**

5 Charging Method Selection: Based on the table below, Type-6 (LEV DC charging) is selected as the most suitable charging method for the proposed commercial three-wheeler goods carrier. It matches the battery architecture, enables fast mid-shift charging, and minimizes downtime during long operating hours. Its simplicity, low cost, and robustness in harsh environmental conditions make it commercially feasible and reliable.

Parameter	Type 6	Type 7	Battery swapping	AC Slow Charging	CCS2
Designed for 3-wheelers	Yes	Yes	Yes	Yes	No
Electrical compatibility	Direct match (60– 120 V DC packs)	Direct match + AC	Pack-specific	Direct match	Mismatch (high-V systems)
Downtime in 10–14 hr shift	Low	Low	Very Low	High	Medium
Daily 100–150	Yes	Yes	Yes	No	Yes

km feasibility					
Payload impact (300–500 kg)	Acceptable	Acceptable	Slight	Acceptable	Not designed
Heat tolerance	High	Medium-high	High	High	Medium
Environmental Protection	High	Medium	Medium	High	Medium
Infra &Operation Cost	Low	Medium	Medium	Least	High
Scalability for fleet	Good	Good	Very Good	Poor	Poor

6 TIRE PRESSURE MANAGEMENT SYSTEM:this system consists of three mechanisms

1. **Automatic pressure sensing and inflation system**
2. **Pressure release mechanism**
3. **Alerting system**

The automatic pressure sensing system operates continuously while the vehicle is in motion. The inflation and pressure release mechanisms, however, are activated only when the vehicle is stationary.
A dedicated pressure sensor is integrated into each tire. Two pressure thresholds are established to ensure safe travel. When the tire pressure falls below the lower bound, the driver is notified to pull over. Inflation commences once the vehicle is stopped and ceases upon reaching the upper pressure bound.
During periods of high ambient temperature, the internal pressure may increase, posing a risk of tire failure. In such events, the driver is immediately alerted, and the excess pressure is safely released through dedicated valves.
A display indicates the pressure level within the reservoir, allowing the driver to ensure it is adequately filled.
For tire punctures, a timer is linked to the wheels. If the inflation system remains active beyond a predetermined duration, pumping is halted, and the tire is flagged as having a puncture.
Low or high tire pressure damages tires early; each tire costs ₹2,500–₹4,000, so preventing early replacement saves money.The system uses a low-cost microcontroller and affordable pressure sensors.
No extra speed sensor is needed because it uses the existing speedometer.
The shared air reservoir and pumping work only when the vehicle is stopped, reducing power use and wear.It avoids expensive temperature-control parts and sealing packs, keeping the overall system cost low
Below is the simplified architecture of the given idea: